

IT REQUEST MANAGEMENT SYSTEM

Vendor Sourcing Brief, Requirements and Solution Blueprint

Prepared for	Client Organisation
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Purpose

This document provides a technology-neutral sourcing brief for vendors to propose, design, build, test, deploy and support a secure web-based IT Request Management System using PHP, Laravel and MySQL. The client organisation name has intentionally been omitted.

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Document Control

Item	Details
Document title	IT Request Management System - Vendor Sourcing Brief, Requirements and Solution Blueprint
Intended audience	Prospective vendors, solution architects, developers, project managers, security reviewers and procurement evaluators
Primary purpose	Support vendor briefing, proposal preparation, technical evaluation and implementation planning
Organisation naming	Generic terms such as Client, Client Organisation and Business Unit are used throughout
Commercial status	This brief is not a contract, purchase order or notice of award. Final scope is subject to procurement and contractual approval.

How Vendors Should Use This Document

- Review the stated business process, functional scope and non-functional requirements.
- Identify assumptions, dependencies, exclusions and proposed alternatives.
- Provide a compliant solution design and explain material deviations.
- Propose an implementation plan, resource model, testing approach and support model.
- Complete a requirements traceability and compliance matrix as part of the proposal.

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1. Executive Summary

The proposed solution is a centralized web application for capturing, reviewing, approving, recommending and tracking IT requests. It replaces fragmented form and email handling with a controlled workflow, structured data, document management, notifications, dashboards and a complete audit trail.

Recommended delivery model

A modular monolith is preferred for the initial release. The application should have clear functional modules and service boundaries while remaining one manageable deployment. Vendors may propose alternatives with justification.

1.1 Target Outcomes

- One authoritative register for IT requests and decisions.
- Standardized submission and validation of mandatory information.
- Transparent approval and recommendation routing.
- Reduced manual follow-up through reminders and escalations.
- Audit-ready records of decisions, comments, documents and timestamps.
- Operational dashboards for workload, aging, bottlenecks and outcomes.

1.2 In-Scope Workflow

Stage	Primary actor	Principal outcome
Draft and submit	Requestor / Project Owner	Complete request with justification, scope, budget, timeline and documents
Owner approval	Project Owner	Approve, reject or return for clarification
Sponsor approval	Project Sponsor	Approve, reject or return for clarification
Completeness review	IT Governance	Validate mandatory information and supporting evidence
Assessment	IT Governance	Assign tier, classification and governance route
Technical recommendations	Relevant IT units	Submit recommendation, conditions and comments
Consolidation	IT HOU / authorized reviewer	Consolidate recommendations and determine route
Committee decision	ITIC or equivalent committee	Decide applicable Full-route requests
Closure	System / Governance	Notify stakeholders, preserve record and close workflow

2. System Requirements

2.1 Proposed Technology Baseline

Layer	Preferred technology	Vendor response expectation
Backend	PHP 8.4+ and current supported Laravel release	Confirm version, support lifecycle and upgrade approach
Database	MySQL 8.4 LTS or approved equivalent	Confirm HA, backup, encryption and recovery design
Frontend	Laravel Blade + Livewire and Tailwind CSS	Demonstrate responsive and accessible user experience
Identity	Microsoft Entra ID using OIDC/OAuth 2.0	Explain SSO, role mapping and account lifecycle
Web tier	Nginx or approved equivalent	Provide secure configuration and hardening
Queue/cache	Redis recommended	Explain deployment, monitoring and failure handling
Source control	Git repository	Define branching, pull request and release controls
Development	Visual Studio Code compatible	Provide reproducible setup and documentation

2.2 User Roles

Role	Core permissions
Requestor	Create, save draft, submit, amend, withdraw where allowed, view own requests
Project Owner	Review and decide assigned requests; provide comments
Project Sponsor	Review and decide assigned requests; provide comments
IT Governance Reviewer	Completeness review, tiering, classification, routing and administration
Technical Reviewer	Provide unit recommendation, conditions and supporting evidence
IT HOU / Consolidator	Consolidate recommendations and determine governance route
Committee Secretariat / Reviewer	Manage committee decision and record conditions
Administrator	Manage users, roles, reference data, templates and configuration
Auditor / Read-only	View authorized records and audit evidence without modification

2.3 Functional Requirements

ID	Area	Requirement
FR-001	Authentication	Users shall sign in using approved enterprise identity and SSO.
FR-002	Profile	The system shall retrieve or maintain requestor department, division and reporting data.
FR-003	Draft	Users shall save incomplete requests as drafts.
FR-004	Submission	The system shall validate mandatory and conditional fields before submission.
FR-005	Identifier	A unique configurable request number shall be generated.
FR-006	Documents	Users shall upload, categorize, preview and download authorized documents.
FR-007	Approval	The system shall support approve, reject and return-for-amendment outcomes.

ID	Area	Requirement
FR-008	Recommendations	Multiple units shall submit independent recommendations without overwriting prior records.
FR-009	Routing	The workflow shall route by tier, classification, decision and governance route.
FR-010	Notifications	The system shall notify users of assignments, decisions, reminders and escalations.
FR-011	Audit	Every material action and transition shall be recorded with actor and timestamp.
FR-012	Reporting	Authorized users shall access filters, dashboards and exports.
FR-013	Administration	Authorized administrators shall manage configurable reference data and templates.
FR-014	Delegation	Authorized temporary delegation shall preserve original and acting approvers.
FR-015	Search	Users shall search by request number, title, status, owner, unit, tier and date.

2.4 Non-Functional Requirements

ID	Category	Requirement
NFR-001	Security	Apply least privilege, secure coding, encryption, secrets management and vulnerability remediation.
NFR-002	Performance	Vendor shall propose measurable response-time and throughput targets based on validated usage volumes.
NFR-003	Availability	Vendor shall propose availability, maintenance and service restoration commitments.
NFR-004	Scalability	Solution shall scale users, requests, documents and workflow volume without redesign of the core domain.
NFR-005	Accessibility	Interfaces shall support keyboard use, readable contrast, labels and validation feedback.
NFR-006	Auditability	Critical records shall be immutable to ordinary users and retained according to approved policy.
NFR-007	Maintainability	Use documented standards, automated tests, modular code and controlled configuration.
NFR-008	Portability	Deployment shall be reproducible across development, test and production environments.
NFR-009	Observability	Provide application logs, security logs, health checks, queue monitoring and alerting.
NFR-010	Data protection	Data collection, access, retention and disposal shall follow Client requirements and applicable obligations.

3. System Design

3.1 Functional Modules

Module	Capabilities
Identity and access	SSO, user profile, roles, permissions, delegation and session handling
Organisation structure	Departments, divisions, heads, reporting relationships and reviewer groups
Request management	Draft, submit, amend, copy, withdraw, view and track
Workflow and approvals	Stage engine, assignments, decisions, due dates and escalation
Recommendation management	Unit reviews, conditions, evidence, versions and consolidation
Document management	Upload, classification, secure retrieval, metadata and malware scanning integration
Notifications	Email/in-app templates, reminders, escalation and delivery status
Reporting	Operational dashboard, aging, turnaround, outcomes and exports
Administration	Reference data, workflow settings, notification templates and user access
Audit and compliance	Event history, change log, evidence export and retention controls

3.2 Workflow State Model

- Draft
- Submitted
- Pending Project Owner
- Pending Project Sponsor
- Pending Completeness Review
- Returned for Amendment
- Pending Technical Recommendation
- Pending Consolidation
- Pending Committee Decision
- Approved
- Approved with Conditions
- Not Recommended / Rejected
- Withdrawn
- Closed

Workflow control

Vendors should implement controlled transitions rather than allowing direct free-text status updates. Each transition must validate authorization, prerequisites and required comments, then create an audit record within the same database transaction.

3.3 Business Rules

Rule ID	Rule
BR-001	Submitted requests cannot be deleted by ordinary users.

Rule ID	Rule
BR-002	A rejection or return decision requires comments.
BR-003	Conditional documents and fields shall be driven by tier and classification.
BR-004	The approver cannot modify the requestor original business justification during approval.
BR-005	Recommendation records shall be versioned or preserved rather than overwritten.
BR-006	Final closure requires all mandatory decisions and documentation.
BR-007	Requests returned for amendment shall resume at the configured stage after resubmission.
BR-008	Delegated decisions shall capture both delegated-from and acting users.
BR-009	System-managed fields shall not be editable through ordinary request screens.
BR-010	All timestamps shall be stored consistently and displayed in the configured local time zone.

4. System Architecture

4.1 Logical Architecture

```

Browser  ->  HTTPS / Reverse Proxy  ->  Laravel Application  ->  MySQL
                                     |-> Redis Queue / Cache
                                     |-> Private File Storage
                                     |-> Entra ID
                                     |-> Email / Notification Service
                                     |-> Monitoring and Log Platform

```

4.2 Application Layers

Layer	Responsibility
Presentation	Responsive screens, forms, dashboards, validation feedback and accessibility
Application	Use cases, workflow commands, authorization, notifications and reporting orchestration
Domain	Request, approval, recommendation and workflow rules
Persistence	Eloquent models, repositories where justified, transactions and database migrations
Integration	Identity, email, storage, scanning, reporting and future API services
Operations	Deployment, configuration, queues, scheduler, logging, monitoring and backup

4.3 Deployment Environments

- Development: local or controlled shared environment with anonymized/non-production data.
- Test/UAT: production-like configuration for integration, security and user acceptance testing.
- Production: hardened environment with controlled release, backup, monitoring and restricted administration.

4.4 API and Integration Principles

- Use versioned REST APIs when external integration is required.
- Use service accounts and managed secrets; do not store credentials in source code.
- Apply input validation, authorization and throttling.
- Record correlation IDs for traceability.
- Design integration failures for retry and reconciliation.
- Document interfaces using OpenAPI or an equivalent standard.

5. Database Design

5.1 Core Data Model

Table	Purpose	Selected key fields
users	Employee and application identity	id, employee_no, name, email, department_id, division_id, manager_id, entra_object_id, is_active
departments	Department reference	id, code, name, head_user_id, is_active
divisions	Division reference	id, code, name, head_user_id, is_active
it_requests	Current request record	request_no, title, requestor_id, owner_id, sponsor_id, tier_id, classification_id, status, current_stage
approval_tasks	One approval assignment or decision	request_id, stage, sequence, approver_id, due_at, decision, comments, decided_at
recommendations	Unit recommendation history	request_id, review_unit_id, reviewer_id, recommendation, conditions, version_no, submitted_at
workflow_histories	Stage transitions	request_id, from_stage, to_stage, action, performed_by, remarks, created_at
attachments	Document metadata	request_id, category, original_name, storage_path, mime_type, size, uploaded_by
comments	User and internal discussion	request_id, user_id, comment_type, comment, is_internal
audit_logs	Before/after change evidence	user_id, auditable_type, auditable_id, event, old_values_json, new_values_json, IP
notifications	Notification tracking	user_id, request_id, template, channel, status, sent_at
reference tables	Configuration values	tiers, classifications, stages, routes, decisions, units, roles, permissions

5.2 Data Design Requirements

- Use foreign keys and indexes for identifiers, user references, status, stage and reporting dates.
- Use decimal data types for financial values and avoid floating-point storage.
- Use soft deletion only where justified; do not permit deletion of audit evidence.
- Keep binary documents outside the relational database unless the hosting standard requires otherwise.
- Record created_by and updated_by where technical audit logging alone is insufficient.
- Define retention and archival rules before production migration.
- Provide data dictionary, ERD and migration scripts as controlled deliverables.

6. UI/UX Design

6.1 Primary Navigation

Menu	Purpose
Dashboard	Summary cards, workload, aging and recent activity
New Request	Guided request wizard
My Requests	Drafts, submissions and returned items
My Approvals	Owner and sponsor decisions
Recommendations	Technical review assignments
Governance Workspace	Completeness, classification, routing and consolidation
Committee Workspace	Committee agenda and decision recording
Reports	Filters, dashboard views and exports
Administration	Users, roles, reference data and templates

6.2 Request Wizard

Step	Contents
1. Request Information	Title, requestor, department, division, owner, sponsor, tier and classification
2. Business Justification	Need, objectives, business-plan alignment, value and impact
3. Budget and Timeline	Amount, source, code, funding type, start, completion and resources
4. Risk and Scope	Urgency, risk, mitigation, dependencies, in-scope and out-of-scope
5. Documents and Review	Supporting documents, declaration, preview and submission

6.3 Experience Requirements

- Responsive desktop and mobile layout.
- Autosave draft or clearly visible Save Draft action.
- Visible progress indicator and section completion status.
- Conditional questions based on selected tier, classification and urgency.
- Inline validation with plain-language correction guidance.
- Status timeline showing completed, current and future stages.
- Accessible controls, keyboard navigation and adequate contrast.
- Clear separation between applicant content and internal governance content.
- Confirmation before irreversible actions.
- Consistent status badges and decision terminology.

7. Development Blueprint

7.1 Suggested Laravel Structure

```

app/
  Actions/
  Enums/
  Events/
  Http/Controllers/
  Http/Requests/
  Jobs/
  Listeners/
  Models/
  Notifications/
  Policies/
  Services/
    RequestService.php
    WorkflowService.php
    ApprovalService.php
    RecommendationService.php
    AuditService.php
  Support/

```

7.2 Engineering Principles

- Keep controllers thin and place business logic in services/actions.
- Use Form Request classes for validation.
- Use Policies and Gates for authorization.
- Use enums or controlled reference data for stages and decisions.
- Wrap submission and workflow transitions in database transactions.
- Use queued notifications and idempotent jobs.
- Write feature tests for each allowed and prohibited transition.
- Use migrations and seeders for repeatable deployment.
- Apply dependency and static analysis checks in CI.
- Maintain technical documentation alongside the source repository.

7.3 Minimum Deliverables

Category	Vendor deliverable
Requirements	Validated requirements, assumptions and traceability matrix
Design	System design, architecture, ERD, data dictionary, security design and interface design
Build	Source code, migrations, configuration, automated tests and build instructions
Testing	Test plan, scripts, evidence, defect register and UAT support
Deployment	Deployment plan, rollback plan, release notes and environment configuration
Operations	Runbook, backup/recovery procedures, monitoring guide and support model
Training	Administrator guide, user guide and knowledge-transfer material
Handover	Repository, credentials transfer process, licences, third-party inventory and acceptance pack

8. Project Plan

Planning note

Vendors shall propose a detailed schedule with duration, dependencies, milestones, resource loading and critical path. The sequence below defines expected phases but does not prescribe commercial duration.

Phase	Key activities	Exit criteria
1. Mobilisation	Kick-off, governance, access, plan and risk register	Approved project management plan
2. Discovery	Workshops, process validation, data and integration assessment	Requirements baseline approved
3. Solution Design	Architecture, UX, ERD, security, interfaces and test strategy	Design approval
4. Foundation Build	Repository, environments, SSO, roles, CI/CD and organisation data	Foundation demonstrated
5. Request Module	Form, drafts, validation, attachments and submission	Request submission accepted
6. Workflow Module	Approvals, amendment, routing, decisions and notifications	Core workflow accepted
7. Governance Module	Completeness, recommendations, consolidation and committee decision	End-to-end process accepted
8. Reporting and Admin	Dashboards, exports, configuration and audit views	Operational reporting accepted
9. Testing	Functional, integration, security, performance and UAT	Acceptance criteria met
10. Deployment	Cutover, rollback readiness, go-live and handover	Production acceptance
11. Stabilisation	Monitoring, defect correction and knowledge transfer	Operational handover completed

8.1 Governance and Reporting

- Weekly progress report covering achievements, plan, risks, issues, decisions and changes.
- Maintained RAID log and decision log.
- Formal design, test, deployment and acceptance checkpoints.
- Change control for scope, schedule, cost and technical baseline.
- Documented action owners and due dates.

8.2 Vendor Team

Role	Expected responsibility
Project Manager	Plan, governance, reporting, risk and stakeholder coordination
Business Analyst	Requirements, process mapping and traceability
Solution Architect	Architecture, integration, security and design assurance
UI/UX Designer	Research, wireframes, prototypes and accessibility
Laravel Developers	Application development, integration and automated tests
Database Engineer	Schema, performance, backup and migration
QA/Test Lead	Strategy, scripts, evidence and defect management

Role	Expected responsibility
DevOps Engineer	Environments, CI/CD, deployment, monitoring and rollback
Security Specialist	Threat modelling, secure configuration and remediation
Trainer/Support Lead	Guides, training, handover and early-life support

9. Testing and Acceptance

9.1 Testing Scope

Test type	Minimum coverage
Unit testing	Domain rules, services, validation and utility components
Feature testing	Routes, permissions, form submission and workflow transitions
Integration testing	Identity, email, storage, queues and approved external services
Security testing	Access control, session handling, input handling, dependency and vulnerability checks
Performance testing	Agreed transaction volumes, document upload, reporting and concurrent use
Backup/recovery testing	Database and document recovery with evidence
UAT	Role-based end-to-end scenarios and reporting reconciliation
Regression testing	Automated and manual verification following material fixes

9.2 Illustrative Acceptance Scenarios

Test ID	Scenario
UAT-001	Requestor saves and resumes a draft.
UAT-002	Mandatory and conditional validation prevents incomplete submission.
UAT-003	Unique request number is generated without duplication.
UAT-004	Project Owner approves and workflow advances.
UAT-005	Project Sponsor rejects with mandatory comments.
UAT-006	Governance returns an incomplete request for amendment.
UAT-007	Resubmitted request resumes at the correct stage.
UAT-008	Multiple technical units submit separate recommendations.
UAT-009	Consolidator records conditions and governance route.
UAT-010	Full-route request reaches committee decision stage.
UAT-011	Unauthorized user cannot view or decide a restricted request.
UAT-012	Audit trail contains actor, timestamp, transition and comments.
UAT-013	Reminder and escalation are triggered according to configuration.
UAT-014	Dashboard totals reconcile with transactional records.
UAT-015	Backup is restored and validated in a controlled test.

10. Security, Operations and Support

10.1 Security Requirements

- Enterprise SSO and role-based access control.
- Least-privilege administration and separation of duties.
- Server-side authorization for every sensitive action.
- Encryption in transit and at rest where supported by the approved platform.
- Secure secret storage and rotation.
- File type, size and malware controls for uploads.
- Protection against common web application vulnerabilities.
- Dependency inventory and patching process.
- Security and audit log integration.
- Documented vulnerability remediation before production acceptance.

10.2 Operational Requirements

- Health endpoint and service monitors.
- Centralized application, queue, scheduler and security logs.
- Database and document backup with tested restoration.
- Documented incident, support and escalation procedures.
- Controlled deployment and rollback.
- Capacity, storage and database health monitoring.
- Configurable retention and archival.
- Administrative runbook and knowledge transfer.

10.3 Warranty and Support

Vendors shall propose warranty, stabilization and ongoing support arrangements, including service coverage, incident priority definitions, response/restoration targets, patching, monitoring responsibilities, release management, escalation contacts and optional enhancement support. Commercial values and final service levels will be agreed through the procurement process.

11. Vendor Response Requirements

The vendor proposal should be structured so that technical and commercial evaluation can be performed consistently.

Section	Required response
Executive response	Understanding of objectives, proposed solution and differentiators
Compliance matrix	Comply / partially comply / not comply, explanation and reference
Solution architecture	Logical, deployment, security, integration and operations design
Implementation methodology	Phases, deliverables, milestones, dependencies and acceptance
Project team	Named or proposed roles, relevant experience and allocation
Testing and quality	Automation, UAT, security, performance and defect management
Data and migration	Reference data setup, any migration method, validation and reconciliation
Security and privacy	Controls, tooling, certifications where applicable and evidence approach
Support model	Warranty, SLA proposal, escalation, maintenance and enhancement options
Commercial response	Transparent pricing assumptions, licences, third-party costs and options
Risks and assumptions	Constraints, dependencies, exclusions and Client responsibilities
Demonstration	Proposed prototype/demo covering key user journeys and audit history

11.1 Mandatory Vendor Declarations

- Identify all third-party and open-source components and applicable licences.
- Declare subcontractors and hosting dependencies.
- Confirm source-code and documentation handover arrangements.
- Declare any material deviation from the preferred technology stack.
- State data-location, access and support-location assumptions.
- Confirm secure development and vulnerability management practices.
- Identify all information required from the Client to finalize sizing and schedule.

12. Appendices

Appendix A - High-Level Field Catalogue

Group	Fields
Request identity	Request number, title, request date, requestor, department, division
Stakeholders	Project Owner, Project Sponsor, reviewers and committee roles
Classification	Tier, classification, governance route and current stage
Business case	Need, objectives, alignment, reference, value and impact
Financial	Budget amount, source, code and funding type
Risk	Urgency, justification, risk, mitigation, dependencies and constraints
Scope and plan	In scope, out of scope, start, completion and resources
Decision	Approval outcomes, recommendations, conditions, dates and comments
System control	Status, timestamps, assignment, SLA, version and audit metadata

Appendix B - Proposal Compliance Matrix Template

Req. ID	Requirement summary	Compliance	Vendor response / evidence
FR-___		Comply / Partial / No	
NFR-___		Comply / Partial / No	
SEC-___		Comply / Partial / No	
DEL-___		Comply / Partial / No	

Appendix C - Assumptions to Validate During Discovery

- Expected user population, concurrent usage and annual request volume.
- Hosting platform, network zones and environment standards.
- Identity attributes available from Entra ID, including department and division mapping.
- Email and notification services.
- Approved file storage, antivirus scanning and document-size limits.
- Retention schedule and audit evidence requirements.
- Committee operating model and whether decisions require voting or only recording.
- Required operational dashboards and report export formats.
- Historical data migration scope.
- Required integrations with service management, procurement, finance or project-governance systems.

Appendix D - Glossary

Term	Meaning
Client	The organisation procuring or sponsoring the solution
IT Request	A formal request for a new system, enhancement, subscription/licence, partnership or other IT-related initiative

Term	Meaning
IT Governance	The function responsible for completeness, classification, routing and governance oversight
IT HOU	Authorized Head of Unit or equivalent consolidating authority
ITIC	IT Investment Committee or equivalent decision body
UAT	User Acceptance Testing
SSO	Single Sign-On
OIDC	OpenID Connect
RBAC	Role-Based Access Control
RAID	Risks, Assumptions, Issues and Dependencies